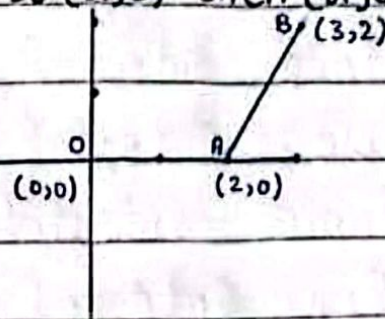


Line Integral: $\int \vec{A} \cdot d\vec{r}$

$\vec{A} = (2x+y)\hat{i} + (3y-x)\hat{j}$, Evaluate $\int \vec{A} \cdot d\vec{r}$

where C is in xy -plane consisting of line segment from $(0,0)$ to $(2,0)$ then $(2,0)$ to $(3,2)$

Solve:



$$\vec{A} = (2x+y)\hat{i} + (3y-x)\hat{j}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\int_C \vec{A} \cdot d\vec{r} = \int_C (2x+y)\hat{i} + (3y-x)\hat{j} \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_C (2x+y)dx + (3y-x)dy$$

$$m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{0 - 2}{2 - 3}$$

Put $y=0$, $dy=0$

$$= \int_0^2 2x dx$$
$$= 2 \left| \frac{x^2}{2} \right|_0^2$$

$$m = 2$$

Equ. of line:

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 2(x - 2)$$

$$y = 2x - 4$$

$$= (2)^2 - 0 = [4] = \int_C \vec{A} \cdot d\vec{r}$$

$$\int_C \vec{A} \cdot d\vec{r} = (2x+y)dx + (3y-x)dy$$

Putting the values of y, dy

$$\int_2^3 [(4x-4) + (5x-12)2] dx$$

$$dy = 2dx$$

$$= \int_2^3 (14x - 28) dx$$

$$= \left| \frac{14x^2}{2} - 28x \right|_2^3 \Rightarrow ((7(3)^2 - 28(3)) - (7(2)^2 - 28(2)))$$

$$\int_C \vec{A} \cdot d\vec{r} = 7$$

$$\int \vec{A} \cdot d\vec{r} = \int_C \vec{A} \cdot d\vec{r} + \int_C \vec{A} \cdot d\vec{r}$$

$$= 4 + 7$$

$$\int \vec{A} \cdot d\vec{r} = 11$$

Vector Analysis:

Vector Integration:

* Line integral is generalization of definite integral.

* Surface integral is generalization of Double integral.

* Volume integral is generalization of Triple integral.

Properties:

$$(i) \int_C k \vec{A} \cdot d\vec{r} = k \int_C \vec{A} \cdot d\vec{r}$$

$$(ii) \int_C (\vec{A} + \vec{B}) \cdot d\vec{r} = \int_C \vec{A} \cdot d\vec{r} + \int_C \vec{B} \cdot d\vec{r}$$

$$(iii) \int_C \vec{A} \cdot d\vec{r} = \int_{C_1} \vec{A} \cdot d\vec{r} + \int_{C_2} \vec{A} \cdot d\vec{r}$$

Q# $\vec{A} = 3xy\hat{i} - y^2\hat{j}$ Evaluate $\int_C \vec{A} \cdot d\vec{r}$, where C is the curve in xy -plane $y = 2x^2$ from $(0,0)$ to $(1,2)$.

Solve:

$$\text{Let } \vec{r} = x\hat{i} + y\hat{j} \Rightarrow d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\vec{A} \cdot d\vec{r} = (3xy\hat{i} - y^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$= 3xy dx - y^2 dy$$

$$\int \vec{A} \cdot d\vec{r} = \int 3xy dx - \int y^2 dy$$

Put y, dy

$$= \int_0^1 3x(2x^2) dx - \int_0^1 (2x^2)^2 (4x dx)$$

$$= \int_0^1 6x^3 dx - \int_0^1 16x^5 dx$$

$$= 6 \left| \frac{x^4}{4} \right|_0^1 - 16 \left| \frac{x^6}{6} \right|_0^1 \Rightarrow \frac{3}{2} (1-0) - \frac{8}{3} (1-0)$$

$$= \frac{3}{2}(1) - \frac{8}{3}(1) \Rightarrow \boxed{\int_0^1 \vec{A} \cdot d\vec{r} = -\frac{7}{6}}$$

4. If $\vec{A} = 2xy^2\hat{i} + 2(x^2y + y)\hat{j}$, Evaluate $\int \vec{A} \cdot d\vec{r}$ from $(0,0)$ to $(2,4)$ along the following:

(i) Straight line $y = 2x$

(ii) Parabola $y = x^2$

$$\text{Let } \vec{r} = x\hat{i} + y\hat{j} \Rightarrow d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\vec{A} \cdot d\vec{r} = (2xy^2\hat{i} + 2(x^2y + y)\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$\int \vec{A} \cdot d\vec{r} = \int 2xy^2 dx + \int 2(x^2y + y) dy$$

$$(i) \quad y = 2x$$

$$dy = 2dx$$

$$= \int_0^2 2x(2x)^2 dx + \int_0^2 2(x^2(2x) + 2x) 2dx$$

$$= \int_0^2 8x^3 dx + \int_0^2 8x^3 dx + \int_0^2 8x dx$$

$$8 \left| \frac{x^4}{4} \right|_0^2 + 8 \left| \frac{x^4}{4} \right|_0^2 + 8 \left| \frac{x^2}{2} \right|_0^2$$

$$\frac{8}{4} (2^4 - 0) + \frac{8}{4} (2^4 - 0) + \frac{8}{2} (2^2 - 0)$$

$$2(16) + 2(16) + 4(4)$$

$$= 32 + 32 + 16$$

$$= 80$$

$$\boxed{\int \vec{A} \cdot d\vec{r} = 80}$$

$$(ii) \quad y = x^2$$

$$dy = 2x dx$$

$$\int \vec{A} \cdot d\vec{r} = \int 2x(x^2)^2 dx +$$

$$\int 2(x^2(x^2) + x^2) 2x dx$$

$$= \int_0^2 2x^5 dx + \int_0^2 4x(x^4 + x^2) dx$$

$$= \int_0^2 2x^5 dx + \int_0^2 4x^5 dx + \int_0^2 4x^3 dx$$

$$= 2 \left| \frac{x^6}{6} \right|_0^2 + 4 \left| \frac{x^6}{6} \right|_0^2 + 4 \left| \frac{x^4}{4} \right|_0^2$$

$$= \frac{2}{6} (2^6 - 0) + \frac{4}{6} (2^6 - 0) + \frac{4}{4} (2^4 - 0)$$

$$= \frac{1}{3} (64) + \frac{2}{3} (64) + 16$$

$$\boxed{\int \vec{A} \cdot d\vec{r} = 80}$$

Theorem: Prove that necessary and sufficient condition for $\int \vec{A} \cdot d\vec{r}$ to be independent of path OR $\vec{A}$ to be conservative in region iff $\oint \vec{A} \cdot d\vec{r} = 0 \quad \forall$ closed path.

Proof: Condition is necessary

Let $\vec{A}$ be conservative

in region so

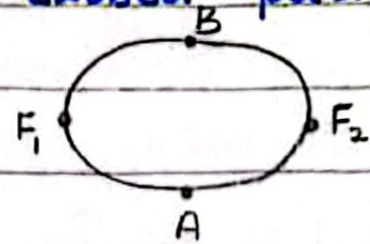
$$\int_{F_1 A F_2} \vec{A} \cdot d\vec{r} = \int_{F_1 B F_2} \vec{A} \cdot d\vec{r}$$

$$\int_{F_1 A F_2} \vec{A} \cdot d\vec{r} - \int_{F_1 B F_2} \vec{A} \cdot d\vec{r} = 0$$

$$\int_{F_1 A F_2} \vec{A} \cdot d\vec{r} + \int_{F_2 B F_1} \vec{A} \cdot d\vec{r} = 0$$

$$\int_{F_1 A F_2 B} \vec{A} \cdot d\vec{r} = 0$$

$$\oint \vec{A} \cdot d\vec{r} = 0 \quad (\text{Proved})$$



Condition is Sufficient

Suppose $\oint \vec{A} \cdot d\vec{r} = 0$

$$\int_{F_1 A F_2 B} \vec{A} \cdot d\vec{r} = 0$$

$$\int_{F_1 A F_2} \vec{A} \cdot d\vec{r} + \int_{F_2 B F_1} \vec{A} \cdot d\vec{r} = 0$$

$$\int_{F_1 A F_2} \vec{A} \cdot d\vec{r} - \int_{F_2 B F_1} \vec{A} \cdot d\vec{r} = 0$$

$$\int_{F_1 A F_2} \vec{A} \cdot d\vec{r} = \int_{F_2 B F_1} \vec{A} \cdot d\vec{r}$$

which shows

$\vec{A}$ is conservative.

Surface Integral:

$$\iint_S \vec{A} \cdot \hat{n} ds = \iint_R \vec{A} \cdot \hat{n} \frac{dx dy}{|\hat{n} \cdot \hat{k}|} \quad \text{for } xy\text{-plane}$$

$$= \iint_R \vec{A} \cdot \hat{n} \frac{dy dz}{|\hat{n} \cdot \hat{j}|} \quad \text{for } yz\text{-plane}$$

$$= \iint_R \vec{A} \cdot \hat{n} \frac{dx dz}{|\hat{n} \cdot \hat{i}|} \quad \text{for } xz\text{-plane}$$

$\vec{F} = 3x\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ at point $(0,0,0)$
to $(2,1,3)$. Find $\int_C \vec{F} \cdot d\vec{r}$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \Rightarrow d\vec{r} = x\hat{i}dx + y\hat{j}dy + z\hat{k}dz$$

$$\vec{F} \cdot d\vec{r} = (3x\hat{i} + (2xz - y)\hat{j} + z\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\vec{F} \cdot d\vec{r} = 3xdx + (2xz - y)dy + zdz$$

We know that

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1} = t$$

$$\frac{x-0}{2-0} = \frac{y-0}{1-0} = \frac{z-0}{3-0} = t$$

$$x/2 = t, y = t, z/3 = t$$

$$x = 2t, y = t, z = 3t$$

$$d\vec{r} = \int 3xdx + \int (2xz - y)dy + \int zdz$$

$$= \int 3(2t)2dt + \int (2(2t)(3t) - t)dt + \int (3t)3dt$$

$$= \int_0^1 12t dt + \int_0^1 12t^2 dt - \int_0^1 t dt + \int_0^1 9t dt$$

$$= 12 \left[\frac{t^2}{2} \right]_0^1 + 12 \left[\frac{t^3}{3} \right]_0^1 - \left[\frac{t^2}{2} \right]_0^1 + 9 \left[\frac{t^2}{2} \right]_0^1$$

$$= \frac{12}{2} ((1)^2 - 0) + \frac{12}{3} ((1)^3 - 0) - \frac{1}{2} ((1)^2 - 0) + \frac{9}{2}$$

$$= 6 + 4 - \frac{1}{2} + \frac{9}{2}$$

$$= \frac{12 + 8 - 1 + 9}{2} = \frac{28}{2}$$

$$\boxed{\int_C \vec{F} \cdot d\vec{r} = 14}$$

If vector field $\vec{A}$ is conservative then

① $\int_P^Q \vec{A} \cdot d\vec{r}$ is independent of path.

② $\oint \vec{A} \cdot d\vec{r} = 0$ for all closed path.

③ $\vec{\nabla} \times \vec{A} = 0$

④ $\vec{A} = \vec{\nabla} \phi$

⑤ $\vec{A} \cdot d\vec{r} = A_1 dx + A_2 dy + A_3 dz$

Evaluate $\iint_S \vec{A} \cdot \vec{n} \, ds$ where $\vec{A} = 18z\hat{i} - 12\hat{j} + 3y\hat{k}$

and S is part of plane $2x + 3y + 6z = 12$ which is located in I^{st} octant.

Solve:

$$\iint_S \vec{A} \cdot \vec{n} \, ds = \iint_R \vec{A} \cdot \vec{n} \frac{dx dy}{|\vec{n} \cdot \hat{k}|} \quad \text{--- (i)}$$

Let

$$\phi = 2x + 3y + 6z - 12$$

$$\vec{\nabla} \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

$$\vec{\nabla} \phi = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$$|\vec{\nabla} \phi| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49}$$

$$|\vec{\nabla} \phi| = 7$$

$$\vec{n} = \frac{\vec{\nabla} \phi}{|\vec{\nabla} \phi|} = \frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{7}$$

$$\vec{n} = \frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}$$

Now

$$\begin{aligned} \vec{A} \cdot \vec{n} &= (18z\hat{i} - 12\hat{j} + 3y\hat{k}) \cdot \left(\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k} \right) \\ &= \frac{36z - 36 + 18y}{7} \end{aligned}$$

Put the value of z

$$= \frac{36 \left[\frac{12 - 2x - 3y}{6} \right] - 36 + 18y}{7}$$

$$= \frac{6(12 - 2x - 3y) - 36 + 18y}{7} = \frac{72 - 12x - 18y - 36 + 18y}{7}$$

$$\vec{A} \cdot \vec{n} = \frac{1}{7} (36 - 12x)$$

$$= \vec{n} \cdot \hat{k}$$

$$= \frac{6}{7}$$

Put the values in (i)

$$\iint_S \vec{A} \cdot \vec{n} \, ds = \iint \frac{1}{x} (36 - 12x) \frac{dx \, dy}{6/x}$$

$$= \frac{1}{6} \int_0^4 \int_0^{\frac{12-3y}{2}} (36 - 12x) \, dx \, dy$$

$$= \frac{1}{6} \int_0^4 \left| 36x - \frac{12x^2}{2} \right|_0^{\frac{12-3y}{2}} dy$$

$$= \frac{1}{6} \int_0^4 \left[36 \left(\frac{12-3y}{2} \right) - \frac{3}{2} \left(\frac{12-3y}{2} \right)^2 \right] dy$$

$$= \frac{1}{6} \int_0^4 \frac{36(12-3y) - 3(144 + 9y^2 - 72y)}{2} dy$$

$$= \frac{1}{12} \int_0^4 (432 - 108y - 432 - 27y^2 + 216y) dy$$

$$= \frac{1}{12} \int_0^4 (108y - 27y^2) dy \Rightarrow \frac{1}{12} \left| 108 \frac{y^2}{2} - 27 \frac{y^3}{3} \right|_0^4$$

$$\frac{1}{12} (54(4^2 - 0) - 9(4^3 - 0)) \Rightarrow \boxed{\iint_S \vec{A} \cdot \vec{n} \, ds = 24 \text{ Ans.}}$$

$$2x + 3y + 6z = 12$$

$$\text{Put } z = 0$$

$$2x = 12 - 3y$$

$$\boxed{x = \frac{12-3y}{2}}$$

$$\text{Put } x = 0$$

$$12 - 3y = 0$$

$$\boxed{y = 4}$$

$$\text{Gradient of } \vec{f} = \vec{\nabla} f$$

$$\text{Divergence of } \vec{f} = \vec{\nabla} \cdot \vec{f}$$

$$\text{Curl of } \vec{f} = \vec{\nabla} \times \vec{f}$$

Find work done of force field.

$$\vec{F} = x^2\hat{i} - xy\hat{j}, \quad \vec{r} = \cos t\hat{i} + \sin t\hat{j} \quad 0 \leq t \leq \pi/2$$

$$\vec{F} = (\cos t)^2\hat{i} - \cos t \sin t\hat{j}, \quad d\vec{r} = (-\sin t\hat{i} + \cos t\hat{j}) dt$$

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} (\cos^2 t\hat{i} - \cos t \sin t\hat{j}) \cdot (-\sin t\hat{i} + \cos t\hat{j}) dt$$

$$= \int_0^{\pi/2} (-\cos^2 t \sin t - \cos^2 t \sin t) dt$$

$$= \int_0^{\pi/2} (-2 \cos^2 t \sin t) dt \Rightarrow 2 \int_0^{\pi/2} \cos^2 t (-\sin t) dt$$

$$= 2 \left| \frac{\cos^3 t}{3} \right|_0^{\pi/2} \Rightarrow \frac{2}{3} (\cos^3(\frac{\pi}{2}) - \cos^3(0))$$

$$= \frac{2}{3} (0 - 1) \Rightarrow -\frac{2}{3} = -\frac{2}{3}$$

Check Conservative

$$\vec{F} = (x-y)\hat{i} + (x-2)\hat{j}$$

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j}$$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ x-y & x-2 & 0 \end{vmatrix}$$

$$= [(1-0) - (0-1)]\hat{k} \\ = [1-0-0+1]\hat{k} \\ = 2\hat{k}$$

$$= 0\hat{i} - 0\hat{j} + \hat{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ x-y & x-2 \end{vmatrix}$$

$$\vec{\nabla} \times \vec{F} = 2\hat{k} \neq 0$$

So $\vec{F}$ is not

$$= \left[\frac{\partial}{\partial x} (x-2) - \frac{\partial}{\partial y} (x-y) \right] \hat{k}$$

conservative.

$$\text{Curl of } \vec{f} = \vec{\nabla} \times \vec{f}$$

$$\vec{\nabla} \times \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ xe^{xy} & 0 & 0 \end{vmatrix}$$

$$= 0\hat{i} - 0\hat{j} + \hat{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ xe^{xy} & 0 \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial x} (0) - \frac{\partial}{\partial y} (xe^{xy}) \right) \hat{k}$$

$$= (0 - x^2 e^{xy}) \hat{k}$$

$$\text{Put } x=1, y=2$$

$$\text{Curl of } \vec{f} = -e^2 \hat{k}$$

$$\text{Curl of } \vec{f} = \vec{\nabla} \times \vec{f}$$

$$\vec{\nabla} \times \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & 0 \\ x^2 & y & 0 \end{vmatrix}$$

$$= 0\hat{i} - 0\hat{j} + \hat{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ x^2 & y \end{vmatrix}$$

$$= \frac{\partial}{\partial x} (y) - \frac{\partial}{\partial y} (x^2)$$

$$= 0\hat{k}$$

$$\boxed{\text{Curl of } \vec{f} = 0}$$

① Fundamental Theorem of Calculus.

$$\int_a^b F'(x) dx = \left| F(x) \right|_a^b = F(b) - F(a)$$

② Fundamental Theorem of line Integral:

$$\int_c \vec{\nabla} f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

③ Green Theorem:

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_c P dx + Q dy$$

④ - Stokes Theorem:

$$\iint_S \text{Curl } \vec{F} \cdot d\vec{s} = \int_c \vec{F} \cdot d\vec{r}$$

⑤ Divergence Theorem:

$$\iiint_V \text{div } \vec{F} dv = \iint_S \vec{F} \cdot d\vec{s}$$

Find Gradient, Divergence and curl of $\vec{f}$ at $(1,2)$.

$$\vec{f} = xe^{xy}$$

$$\text{Gradient of } \vec{f} = \vec{\nabla} f$$

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$$

$$= \frac{\partial}{\partial x} (xe^{xy}) \hat{i} + \frac{\partial}{\partial y} (xe^{xy}) \hat{j}$$

$$= \left[x \frac{\partial}{\partial x} e^{xy} + e^{xy} \frac{\partial}{\partial x} (x) \right] \hat{i} + x \frac{\partial}{\partial y} e^{xy} \hat{j}$$

$$= \left[x(e^{xy} y) + e^{xy} \right] \hat{i} + x(e^{xy} x) \hat{j}$$

$$= [xye^{xy} + e^{xy}] \hat{i} + x^2 e^{xy} \hat{j}$$

$$= (xy+1)e^{xy} \hat{i} + x^2 e^{xy} \hat{j}$$

$$\text{Put } x=1, y=2$$

$$= 3e^2 \hat{i} + e^2 \hat{j}$$

$$\boxed{\text{Gradient of } \vec{f} = e^2(3\hat{i} + \hat{j})}$$

$$\text{Divergence of } \vec{f} = \vec{\nabla} \cdot \vec{f}$$

$$\vec{\nabla} \cdot \vec{f} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} \right) \cdot (xe^{xy} \hat{i})$$

$$= \frac{\partial}{\partial x} (xe^{xy}) + \frac{\partial}{\partial y} (0)$$

$$= xye^{xy} + e^{xy}$$

$$= e^{xy}(xy+1)$$

$$\text{Put } x=1, y=2$$

$$\text{Divergence of } \vec{f} = e^2(3)$$

$$\vec{\nabla} \cdot \vec{f} = 3e^2$$

$$\vec{f} = x^2 - y$$

$$\text{Gradient of } \vec{f} = \vec{\nabla} f$$

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$$

$$= \frac{\partial}{\partial x} (x^2 - y) \hat{i} + \frac{\partial}{\partial y} (x^2 - y) \hat{j}$$

$$= 2x \hat{i} + (0 - 1) \hat{j}$$

$$= 2x \hat{i} - \hat{j}$$

$$\text{Put } x=1, y=2$$

$$= 2(1) \hat{i} - \hat{j}$$

$$\boxed{\text{Gradient of } \vec{f} = 2\hat{i} - \hat{j}}$$

$$\text{Divergence of } \vec{f} = \vec{\nabla} \cdot \vec{f}$$

$$\vec{\nabla} \cdot \vec{f} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} \right) \cdot (x^2 \hat{i} - y \hat{j})$$

$$= \frac{\partial}{\partial x} (x^2) + \frac{\partial}{\partial y} (-y)$$

$$= 2x + (-1)$$

$$= 2x - 1$$

$$\text{Put } x=1, y=2$$

$$= 2(1) - 1$$

$$= 2 - 1$$

$$\boxed{\text{Divergence of } \vec{f} = 1}$$